

Meshkat Shariat Bagheri

AI Engineer | Agentic AI | Local LLMs | RAG | NLP | OCR

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PROFESSIONAL SUMMARY

AI Engineer with end-to-end experience designing, building, deploying, and maintaining enterprise **agentic AI, local LLM, RAG, NLP, OCR, and ASR systems**. Delivered production and pilot solutions across document intelligence, CRM analytics, natural-language database access, Persian summarization and letter drafting, workflow automation, and large-scale OCR. Experienced in integrating AI systems with OracleDB, Dynamics 365, Exchange/EWS, RocketChat, Zulip, and internal document platforms, with a strong focus on local deployment, access control, evaluation, observability, and reliable structured output.

TECHNICAL SKILLS

- **AI / LLM:** Agentic AI, RAG, Tool Calling, Structured Output, Prompt Engineering, LLM Evaluation, LLMops, Local LLM Serving, Fine-tuning
- **NLP / ML:** NLP, OCR, ASR, Information Retrieval, Semantic Search, Embeddings, Vector Search, Text Summarization, GNNs, Deep Learning, Machine Learning
- **Frameworks:** LangChain, LangGraph, vLLM, PyTorch, Transformers, Ollama, Llama.cpp, FastAPI, Streamlit, Gradio, Django REST Framework
- **Databases:** FAISS, Milvus, Qdrant, OracleDB, PostgreSQL, MySQL, Redis
- **Languages / Tools:** Python, SQL, JavaScript, TypeScript, Go, Git, Docker, Linux, Nginx, Swagger, Jira

PROFESSIONAL EXPERIENCE

AI Engineer

Jun 2025 – Present

Gam Electronics – Almas Team

Tehran, Iran

- Owned the end-to-end delivery of local-first enterprise AI products, from requirements and architecture through prototyping, evaluation, stakeholder demos, deployment, and production maintenance.
- Built a reusable agentic RAG platform with local reasoning models, streamed tool-calling workflows, conversation memory, access control, evaluation hooks, and tracing; integrated it with OracleDB, Dynamics 365, Exchange/EWS, RocketChat, Zulip, and internal document systems.
- Developed an access-controlled document-intelligence platform combining OCR, hybrid dense/BM25 retrieval, Persian exact-match ranking, source reconstruction, per-document authorization, feedback analytics, and incremental indexing across **250k+ OracleDB records**.
- Designed an agentic document-review and fact-checking system for PDF, Word, PowerPoint, HTML, TXT, Markdown, and CSV files, using claim extraction, bidirectional evidence retrieval, NLI, rule-based checks, and LLM validation to identify contradictions and missing evidence.
- Built a natural-language-to-OracleDB search application and improved user satisfaction by nearly **30%** through Qwen3:8B fine-tuning and enhanced structured information extraction.
- Delivered CRM intelligence products for semantic case retrieval, full case-detail loading, quality scoring, expert feedback, aggregate analytics, monthly trends, topic clustering, and Persian/Shamsi metadata repair.
- Developed grounded Persian AI services for administrative letter drafting, revision, replies, and follow-ups, as well as direct and hierarchical long-document summarization, with structured APIs, hallucination safeguards, output-repair loops, authentication, tests, and vLLM-based local inference.
- Built enterprise productivity and speech workflows including meeting-minutes generation, invoice generation, email RAG and follow-up drafting, stand-up summaries, task import, page-aware browser automation, and Persian ASR.
- Automated indexing, observability, analytics, and maintenance through delayed-row lookbacks, OCR/no-OCR jobs, dynamic FAQ generation, local Langfuse tracing, scoring observations, shared evaluation hooks, and clustered usage analysis.
- **Core stack:** *Python, LangChain, LangGraph, vLLM, Ollama, Llama.cpp, PyTorch, Transformers, FastAPI, Milvus, FAISS, SQLite, OracleDB, Dynamics 365, Langfuse, OCR, RAG, LLM fine-tuning.*

Website Administrator

Jan 2023 – Jan 2024

Bio-Ceramic & Implants, University of Tehran

Tehran, Iran

- Managed and updated website content, increasing traffic by 15% and user satisfaction by 20% within 6 months.
- Used HTML, CSS, React.js, Express.js, and Google Analytics to improve content delivery and performance tracking.

Software Engineer

Jun 2023 – Oct 2023

Gam Electronics – Zomorrod Team

Tehran, Iran

- Boosted system performance by 30% and responsiveness by 15% via architecture and feature enhancements.
- **Stack:** .NET, Jira, Swagger, Git

Software Developer

Aug 2019 – Mar 2020

Najva – Retention Marketing Platform

Tehran, Iran

- Improved backend service speed by up to 30% through Python/Django optimization and refactoring.
- Reduced report query time by 25%, improving backend efficiency and user experience.
- Improved code quality by 40% by applying clean-code practices, PEP8, and collaborative review workflows.
- **Stack:** Python, Django REST Framework, Linux, Nginx, Redis, Git, Jira.

PROJECTS

remote-embedding (GitHub, PyPI) | Python, FastAPI, LangChain, Hugging Face, PyPI Apr 2026 – Present

- Published a Python package that exposes a shared FastAPI embedding service and a LangChain-compatible `RemoteEmbeddings` client for RAG applications.
- Enabled multiple applications to reuse one server-side Hugging Face embedding model instance, reducing duplicated VRAM usage on limited GPU hardware.
- Implemented configurable CLI/environment options for model selection, local model cache paths, CPU/GPU device choice, batching, request limits, CUDA cache cleanup, model kwargs, and encode kwargs.
- Added client-side timeout handling, expected-dimension validation, per-request model overrides, HTTP connection cleanup, and context-manager support.

Open-Source Contributor – Google LangExtract (GitHub) | Python, Git Aug 2025 – Present

- Added Llama.cpp support to enable local CPU-based model execution.
- Improved multilingual rendering support for Persian, Chinese, and Japanese.
- Contributed to local-first information extraction workflows for LLM-based applications.

Open-Source Contributor – browser-use WebUI (GitHub) | Python, Git Jan 2025 – Present

- Improved UX for a 55k+ star open-source browser-agent project by adding default LLM configurations and suppressing unnecessary warnings.
- Refactored LLM configuration handling to improve maintainability and reduce setup friction.
- Fixed runtime issues, including `UnboundLocalError` cases affecting browser startup and shutdown.
- Updated CLI documentation for theme-related options.

EDUCATION

University of Tehran

Tehran, Iran

M.Sc. in Information Technology Engineering

Sep 2020 - Sep 2024

- *Thesis Title:* "Text Summarization with Graph Neural Networks;" Introduced a novel *Community Detection Long-Text Summarization (CDLTS)* model combining community detection techniques with GAT.
- *Supervisor:* Dr. Salehi
- *Relevant Coursework:* Text Mining(A+), Machine Learning(A), Social Networks(A+), Complex Networks(A), Information Network Analysis (A)
- *GPA w/o failed courses:* 3.61 (Last two semesters: 3.75)

University of Tehran

Tehran, Iran

B.Sc. in Computer Engineering

Sep 2015 - Sep 2020

- *Thesis Title:* Implementation of the OpenStack Tool as an Internet of Things (IoT) Platform
- *Supervisor:* Dr. Khonsari
- *Relevant Coursework:* AI, Internet Engineering, Data Structure, Algorithm Design, Advanced Programming

PUBLICATIONS

DCOR: Anomaly Detection in Attributed Networks via Dual Contrastive Learning Reconstruction, H. RafieiZade, H. Zare, M. Ghassemi Parsa, H. Davardoust, and **M. Shariat Bagheri**. Complex Networks & Their Applications XIII. COMPLEX NETWORKS 2024 2024. Studies in Computational Intelligence, vol 1189. Springer, Cham. Available at [Springer Chapter](#).

TEACHING EXPERIENCE

University of Tehran

Tehran, Iran

Teaching Assistant

2019 - 2024

- Information Network Analysis (Inst. **Dr. Salehi**) Fall 2024
- Mathematical Theory of Systems (Inst. **Dr. Ebrahimi**) Fall 2024
- Research Methods (Inst. **Dr. Zare**) Fall 2023
- Computer Networks Security (Inst. **Dr. Sayyad Haghighi**) Fall 2019, Spring 2020, Fall 2020
- Operating Systems (Inst. **Dr. Kargahi**) Fall 2019, Spring 2020
- Computer Networks (Inst. **Dr. Khonsari**) Fall 2019, Spring 2020

MISCELLANEOUS EXPERIENCE

Awards and Achievements

- **Master's Entrance Exam** – Among top 1% of candidates in Information Technology national Master's entrance exam 2020
- **CodeCup** – Ranked 257th in the Primary CodeCup Contest among more than 1400+ Teams. 2016
- **Bachelor's Entrance Exam** – Among top 0.1% in the university entrance exam in Computer Engineering national entrance exam. 2015